CIS. 130 - Computer Organization
Assignment 1: Number Systems

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1) Convert Decimal to Binary

a) 17

17
8 R 1
4 R 0
2 R 0
1 R 0
0 R 1

0001 0001

B) 205

205
102 R1
51 R0
25 R1
12 R1
6 R0
3 R0
1 R1
0 R1

1100 1101

C) 337

337
168 R1
84 R0
42 R0
21 R0
10 R1
5 R0
2 R1
1 R0
0 R1

0001 0101 0001

2) Convert 0110 1010 (Binary) to Decimal, Octal, hexadecimal.
128 64 32 16 8 4 2 1

0 1 1 0 1 0 1 0
128 64 32 16 8 4 2 1

64 + 32 + 8 + 2 = 106 - Decimal Value

4 2 1 4 2 1 4 2 1
0 0 1 1 0 1 0 1 0

1 5 2 = 152 octal Value

8 4 2 1 8 4 2 1
0 1 1 0 1 0 1 0

6 $\boxed{10}$ = 6A hexadecimal Value

A = 10
B = 11
C = 12
D = 13
E = 14
F = 15

3) Convert Decimal number 21 to base 3, 5, 8

21 / 3 = 7 R 0
7 / 3 = 2 R 1
2 / 3 = 0 R 2

Base 3 = 210

2 1 0
2×3^2 1×3^1 0×3^0
18 3 0 = 21

21 / 5 = 4 R 1
4 / 5 = 0 R 4

Base 5 = 41

4 1
4×5^1 1×5^0
20 1 = 21

21 / 8 = 2 R 5
2 / 8 = 0 R 2

Base 8 = 25

2 5
2×8^1 5×8^0
16 + 5 = 21

4) Convert 1134 (Base 5 to decimal)

1 1 3 4
1×5^3 1×5^2 3×5^1 4×5^0
125 25 15 4 = 169 decimal Value